Edits in the Wild

A Multilingual Benchmark for Localizing On-Device Generative Image Edits on Consumer Platforms

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Phones now ship generative editors that erase, move and sketch objects into real photos. Today's VLMs can ***agree on what changed*** but ***cannot beat a constant guess at how*** and compression makes them forget, not fabricate.

1 Motivation

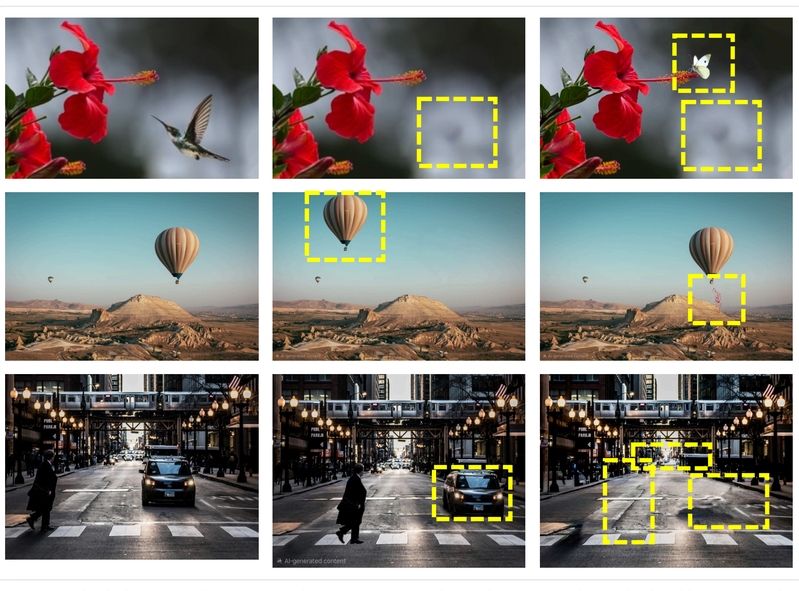
Scale: Apple Intelligence, Galaxy AI and Magic Editor let anyone remove, move or sketch an object into a photo in three taps.

Black boxes: proprietary on-device tools with no API; artifacts differ by platform and device.

Sharing: images are recompressed and re-edited across devices before anyone inspects them.

Gap: existing benchmarks use research models or APIs, English only, and ask "is it synthetic?" — not *what changed, where, and how*.

2 The Benchmark



Originals (left) and edits on an iPhone 16 Pro and a Galaxy S25 Ultra; dashed boxes mark edited regions.

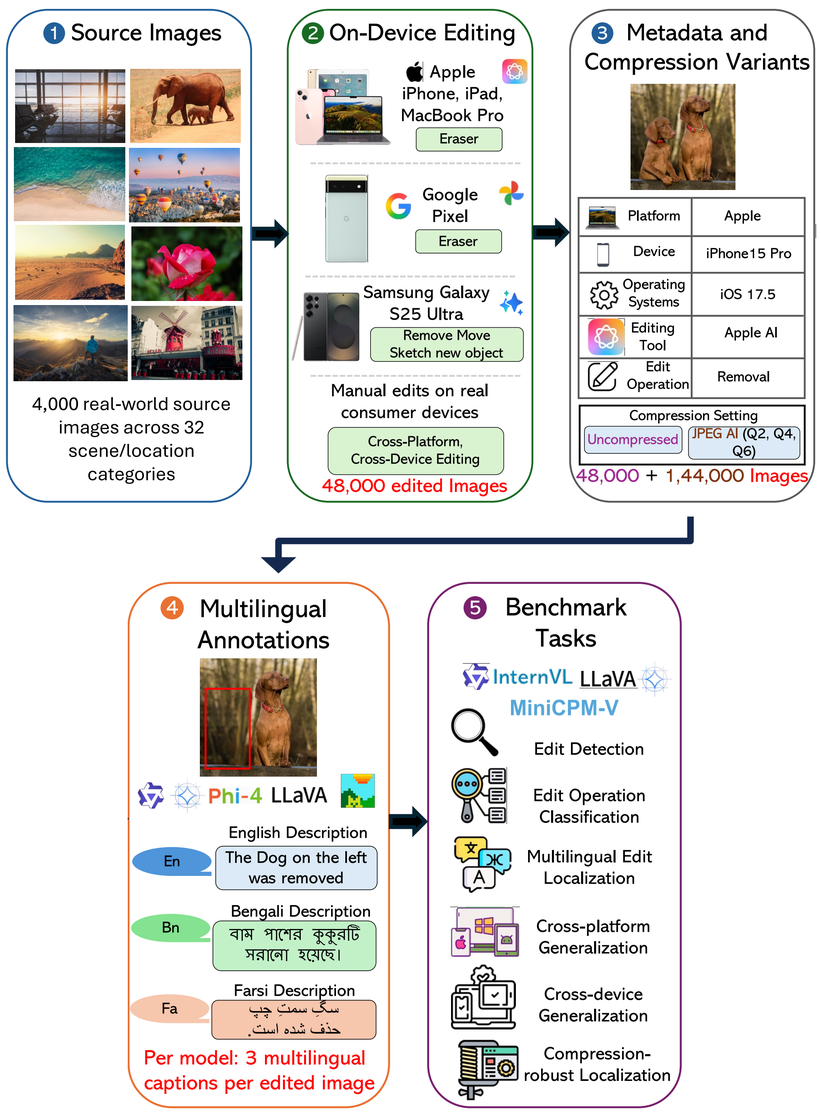
4,000 source photos

48,000 paired on-device edits

144,000 compressed variants

English · বাংলা · فارسی captions in 3 languages

3 How It Is Built



Real devices and native interfaces, no APIs · learned compression at Q6 / Q4 / Q2 · five open VLMs captioning in three languages.

4 Results

Cross-VLM agreement (%)

Model	Qwen3	Gemma	Phi-4	Pixtral	LLaVA	Mean	vs. maj.	LOO κ
Qwen3-VL 32B	100	55.0	58.2	**65.6**	60.6	59.9	69.2	0.201
Gemma 3 27B	55.0	100	**44.0**	54.1	50.4	**50.9**	**56.4**	0.271
Phi-4-RV 15B	58.2	**44.0**	100	58.6	57.3	54.5	60.1	0.239
Pixtral 12B	**65.6**	54.1	58.6	100	61.3	59.9	69.0	0.202
LLaVA-OV 7B	60.6	50.4	57.3	61.3	100	57.4	63.4	0.261

Correctness vs. human references (%)

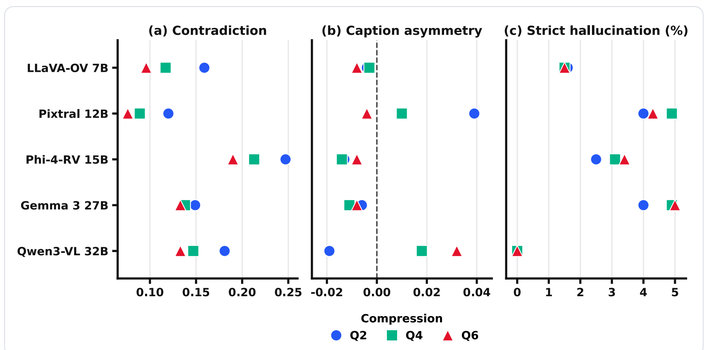
Model	Object (En)	Operation (En)	Full (En)	Full (Bn)	Full (Fa)
Gemma 3 27B	62.8	**55.8**	**35.8**	14.7	21.8
Qwen3-VL 32B	**77.9**	**42.3**	32.7	**39.3**	**35.9**
LLaVA-OV 7B	55.6	**49.9**	28.0	0.4	1.6
Pixtral 12B	44.2	**57.3**	27.8	7.1	12.5
Phi-4-RV 15B	67.1	**34.6**	23.3	13.5	17.5

Majority-class baseline for the operation: **62.2%** · human agreement ceiling for fully correct: **39.6** (En) · **37.2** (Bn) · **36.0** (Fa).

CONSISTENCY VS. CORRECTNESS

Five VLMs reach a **majority label but rarely unanimity** (κ 0.20–0.27). Against human references **none names the operation better than the majority class**. **Qwen3-VL is more often fully correct in Bangla (39.3) and Farsi (35.9) than any model is in English (35.8).**

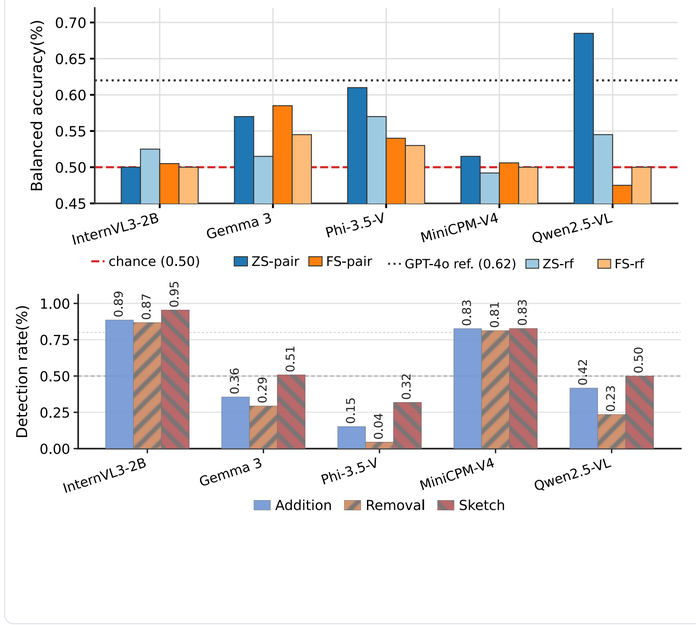
NLI error analysis



OMISSION, NOT FABRICATION

Under compression, contradiction rises but entailment asymmetry stays near zero and strict hallucination stays under 5%: models drop detail rather than invent it.

5 Edit Detection



Balanced accuracy vs. chance (0.50) and GPT-4o (0.62); detection rate by edit type.

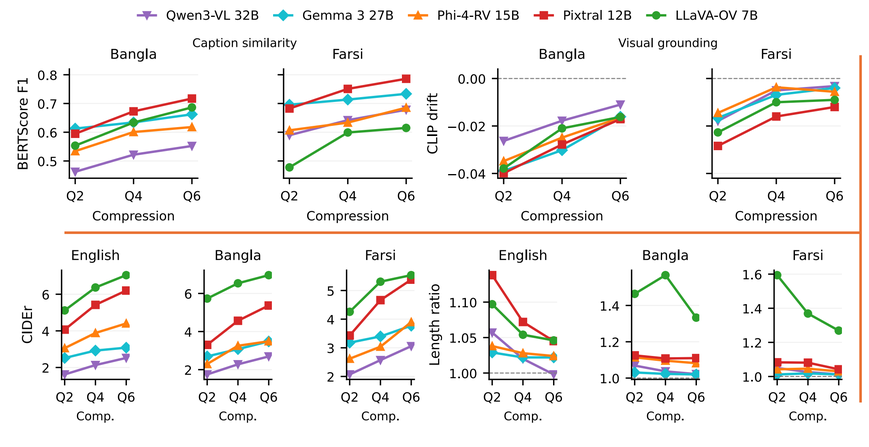
THE ORIGINAL IMAGE IS THE LEVER

Reference-free accuracy sits at chance; paired input lifts only Qwen2.5-VL (0.69) and Phi-3.5-V (0.61); few-shot lowers them. **Removal is the hardest edit to see.**

6 Takeaway

For benchmarks: evaluate consumer edits where they happen — real devices, paired originals, compression tiers, the user's own language — and report cross-VLM agreement rather than trusting one annotator.

For detection: supply the original image, test on removals, and treat compression as lost detail to recover, not a source of hallucination.

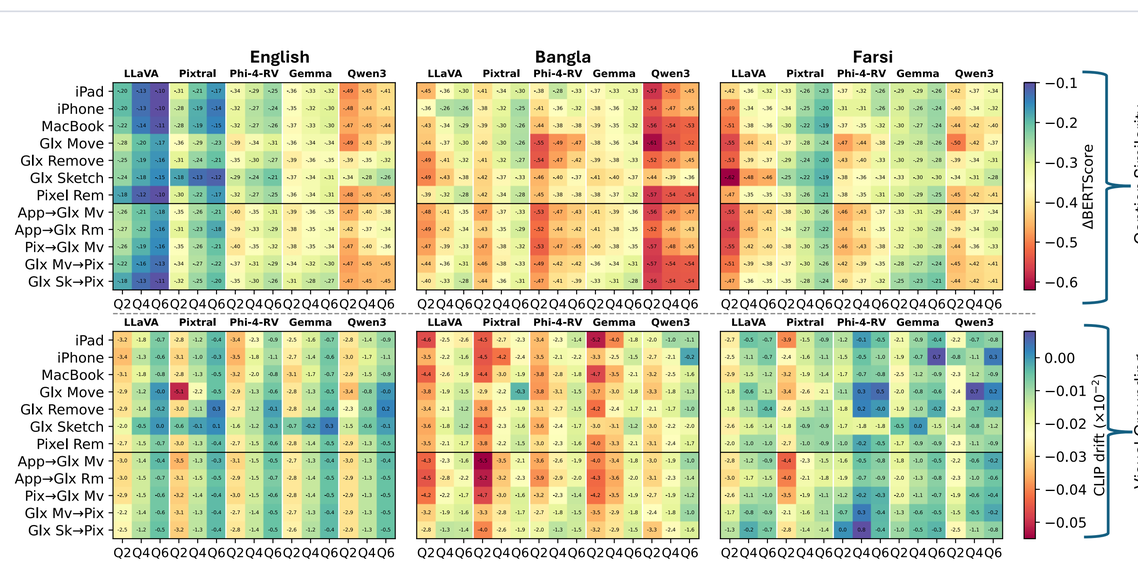


Captioning under compression

Caption similarity, visual grounding, CIDEr and length ratio across Q2–Q6 in English, Bangla and Farsi; right: agreement with English.

WORDS DRIFT, GROUNDING HOLDS

Surface scores fall from Q6 to Q2 in every language while CLIP drift stays near zero. Agreement with English rises with model size, 27% → 74%.



Per-platform effects

ΔBERTScore (top) and CLIP drift (bottom) for twelve editing endpoints, five models, three languages, Q2 / Q4 / Q6.

QUALITY DRIVES THE DRIFT, NOT THE DEVICE

Read by column, not by row: both metrics move toward zero from Q2 to Q6 for every model and language, while single-device edits and cross-device chains stay flat.

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